

EXPERIMENT 8

Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Aim:

To write program for Capacitance of co-axial cable per unit length using SCILAB.

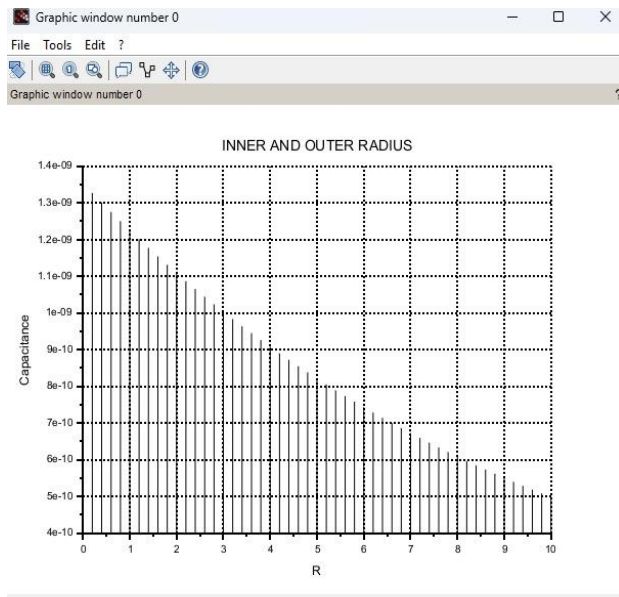
Software required:

- SCILAB version 6.0.1

Program:

```
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
EXP 8.sci
5 // Input-values
6 r1 = input("Enter the value of radius1 (m):-");
7 r2 = input("Enter the value of radius2 (m):-");
8 er = input("Enter the value of relative permittivity:-");
9
10 // Constants
11 e0 = 8.854 * 10^-12;
12
13 // Check validity
14 if r2 <= r1 then
15     disp("Error:-radius2 must be greater than radius1");
16 else
17     // Permittivity
18     e = e0 * er;
19
20     // Capacitance per unit length
21     c = (2 * %pi * e) / log(r2/r1);
22
23     disp("Capacitance (F/m):-");
24     disp(c);
25
26     // Data for graph
27     x = 0:0.2:10;
28     y = c * exp(-0.1 * x);
29
30     // Stem graph
31     plot2d3(x, y);
32
33     // Increase line thickness
34     a = gca();
35     a.thickness = 2;
36
37     xlabel("R");
38     ylabel("Capacitance");
39     xtitle("INNER AND OUTER RADIUS");
40
41     xgrid();
42
43 end
44
45
```

Output:



Practical value:

Enter the value of radius1 (m): 3

Enter the value of radius2 (m): 4

Enter the value of relative permittivity: 7

"Capacitance (F/m) = "
1.354D-09

EXP NO: 08

Determination and plotting of capacitance of coaxial cable per unit length with different (i) inner radius (ii) Outer radius and (iii) relative permittivity of the dielectric medium:

Output:

Enter the radius 1: 3 mm

Enter the radius 2: 4 mm

Enter the value of ϵ_r : 7

Practical output:

"Capacitance per unit length (F/m) = "
1.354D-09

Theoretical calculation:

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

Coaxial cable capacitance per unit length

$$C' = \frac{2\pi\epsilon_0\epsilon_r}{\ln(r_2/r_1)}$$

$r_1 = 3 \times 10^{-3} \text{ m}$

$r_2 = 4 \times 10^{-3} \text{ m}$

$$C' = \frac{2\pi(8.854 \times 10^{-12})(7)}{\ln\left(\frac{4 \times 10^{-3}}{3 \times 10^{-3}}\right)}$$

$$= \frac{2\pi(8.854 \times 10^{-11})(7)}{\ln(4/3)}$$

$\ln(4/3) = 0.28768$

$$C' = \frac{3.893 \times 10^{-10}}{0.28768}$$

$C' = 1.3536 \times 10^{-9} \text{ F/m}$

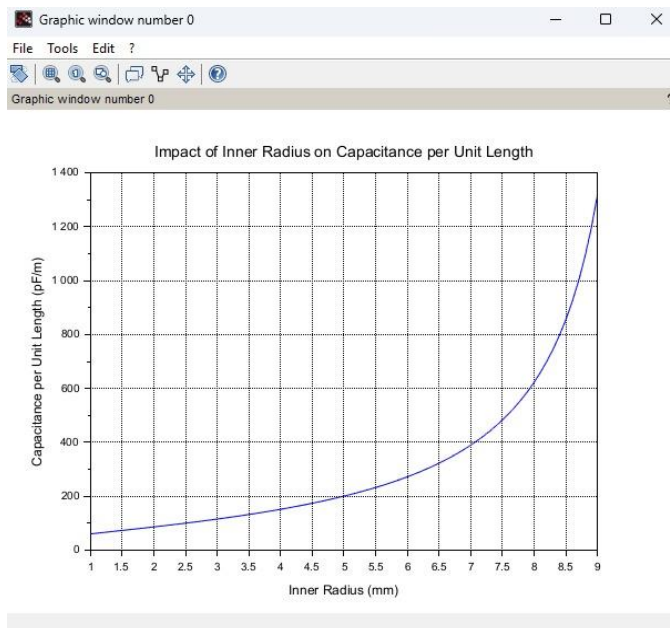
$C' = 1.354 \text{ nF/m}$

CASE STUDY 1: Variation with Inner Radius

Program:

```
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
EXP 8.sci x
1 //Clear previous data
2 clc;
3 clear;
4 clf;
5
6 //Constants
7 epsilon_0 = 8.854e-12; //Permittivity of free space (F/m)
8 epsilon_r = 2.5; //Relative permittivity
9 r2 = 0.01; //Outer radius (m)
10
11 //Inner radius range
12 r1_min = 0.001; //1 mm
13 r1_max = 0.009; //Just below outer radius
14 r1 = r1_min:0.0001:r1_max; //Inner radius values
15
16 //Capacitance per unit length calculation
17 C = (2 * pi * epsilon_0 * epsilon_r) ./ log(r2 ./ r1);
18
19 //Plot
20 plot(r1 * 1000, C * 1e12);
21
22 //Customise plot
23 xlabel("Inner Radius (mm)");
24 ylabel("Capacitance per Unit Length (pF/m)");
25 title("Impact of Inner Radius on Capacitance per Unit Length");
26
27 //Grid
28 xgrid();
29
30 //Display message
31 disp("Plot generated successfully.");
32
```

Output:

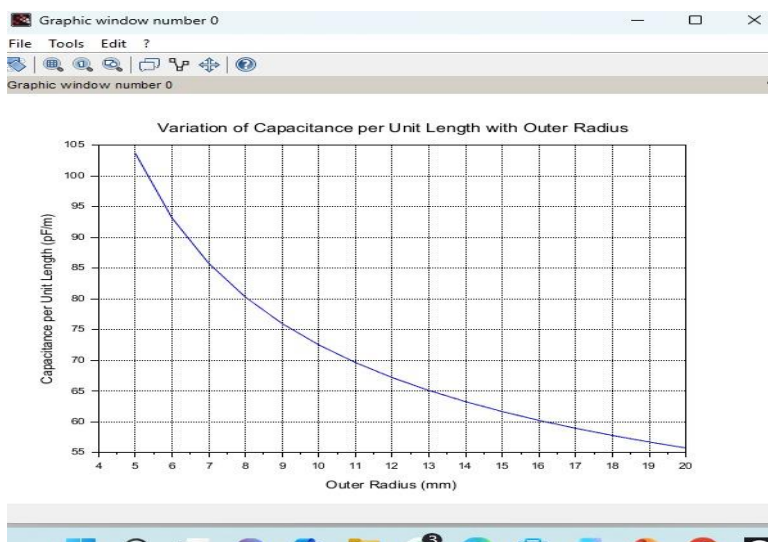


CASE STUDY 2: Variation with Outer Radius

Program:

```
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
EXP 8.sci X
1 //Clear the console and workspace
2 clc;
3 clear;
4 clf;
5
6 //Constants
7 epsilon0 = 8.854e-12; //Permittivity of free space (F/m)
8 epsilon_r = 3.0; //Relative permittivity
9 a = 1e-3; //Inner radius (1 mm)
10
11 //Outer radius values (5 mm to 20 mm)
12 b_values = 5e-3:1e-3:20e-3;
13
14 //Initialize capacitance array
15 C_prime = zeros(b_values);
16
17 //Calculate capacitance per unit length
18 for i = 1:length(b_values)
19     b = b_values(i);
20     epsilon = epsilon0 * epsilon_r;
21     C_prime(i) = (2 * pi * epsilon) / log(b / a);
22 end
23
24 //Plot the results
25 clf(0);
26 plot(b_values * 1e3, C_prime * 1e12);
27
28 //Labels and title
29 xlabel("Outer Radius (mm)");
30 ylabel("Capacitance per Unit Length (pF/m)");
31 title("Variation of Capacitance per Unit Length with Outer Radius");
32
33 //Display grid
34 xgrid();
35
36 //Display message
37 disp("Plot generated successfully.");
38
```

Output:



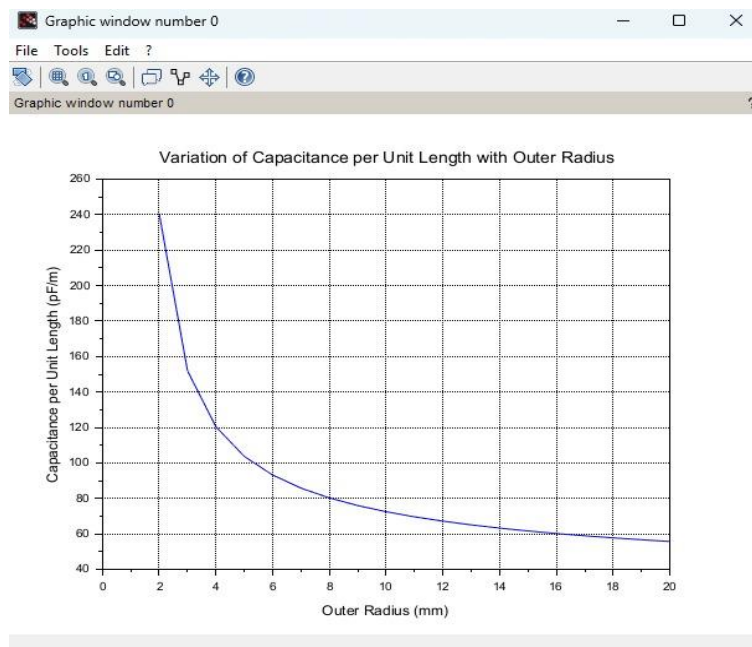
CASE STUDY 3: Variation with Relative Permittivity

Program:

```
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes
EXP 8.sci

1 clear;
2 clear;
3 clf;
4
5 //Constants
6 epsilon0 = 8.854e-12;
7 epsilon_r = 3;
8 a = 1e-3; //Inner radius (1 mm)
9
10 //Outer radius from 1 mm to 20 mm
11 b = 1e-3:1e-3:20e-3;
12
13 C = zeros(b);
14
15 for i = 1:length(b)
16     if b(i) > a then
17         C(i) = (2 * pi * epsilon0 * epsilon_r) / log(b(i)/a);
18     else
19         C(i) = nan; //Ignore b=a
20     end
21 end
22
23 //Plot
24 clf(0);
25 plot(b*1000, C*1e12);
26
27 xlabel("Outer Radius (mm)");
28 ylabel("Capacitance per Unit Length (pF/m)");
29 title("Variation of Capacitance per Unit Length with Outer Radius");
30
31 xgrid();
32
```

Output:



Result :

Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.